

Welcome

Balázs Lengyel

Economic Complexity, Class 1 – Sept 16 2026

Social Data Science, Corvinus University of Budapest

Course focus

- How can we understand economic and technological development through networks and complexity?
- How do **countries, cities, and firms** develop?
- How do **networks** shape economic and technological change?
- Why do economic activities **concentrate in space**?
- How do **agglomeration and specialization** influence development?
- How can we connect **economic theories with network and data science**?

Three perspectives

- **Economic Geography**

Where does economic activity happen — and why?

- **Economic Networks**

How are firms, people, technologies, and places connected?

- **Economic Complexity**

How do capabilities and related activities shape development?

Course structure

- **1. Foundations of Economic Complexity**
Relatedness • Capabilities • Complexity
- **2. Innovation and Economic Geography**
Cities • Agglomeration • Specialization • Networks • Diffusion
- **3. Recent Research**
Inequality • Resilience • Systemic risk • New dimensions of complexity

Week	Date	Topics and details	Moduls	Ref	HW and Quiz
1	Sept 16	Balázs Lengyel: Introduction and overview of the course	I. Foundations of Economic Complexity	[1]	NO
		César Hidalgo: The principle of relatedness			
2	Sept 23	César Hidalgo: Economic complexity	I. Foundations of Economic Complexity	[2]	Quiz1
3	Sept 30	Balázs Lengyel: The development of cities and regions 1: agglomeration, innovation, and technological development	II. Innovation and Economic Geography	[3]	Quiz2
					HW1
4	Oct 7	Balázs Lengyel : The development of cities and regions 2: the specialization-diversity debate		[4]	Quiz3
5	Oct 14	Zoltán Elekes: Labor markets and skills: labor mobility, skill-relatedness		[5,6]	Quiz4
					HW2
6	Oct 21	WRITTEN EXAM 1		[7,8]	NO
		Balázs Lengyel: Learning and innovation in networks: knowledge sourcing and dynamics of networks			
7	Nov 4	Balázs Lengyel and Sándor Juhász: Spatial and network inequalities: social capital, segregation and mixing	II. Innovation and Economic Geography	[9, 10]	Quiz5
					HW3
8	Nov 11	Balázs Lengyel: Spatial diffusion of innovation: agents of change, interregional networks, and spatial life cycles		[11,12]	Quiz6
9	Nov 18	Presentation of students' research proposal			NO
10	Nov 25	Gergő Tóth and Balázs Lengyel: Shocks and adaptation: network resilience, systemic risks	III. Recent research	[13,14]	Quiz7
					HW4
11	Dec 2	Viktor Stojkoski: Digital trade and Multidimensional Economic Complexity	III. Recent research	[15,16]	Quiz8
12	Dec 9	WRITTEN EXAM 2			NO
		OR Presentation of Projects (this will happen outside the class but the same week)			

Evaluation and grading

1. **Class activity (weekly):** Students' activity in the class asking and answering questions. - 3%
2. **Literature (weekly):** Each class will include a discussion of the assigned reading, which will be evaluated by literature quizzes. - 24%
3. **Homework (biweekly):** Students will collect short case studies, and work on other types of short assignments related to the class material. - 13%
4. **Research Proposals** in groups of 2: proposals will include a research problem, research question, data sources, methodology, expected results, and impact presented on week 9. - 30%
5. **Written Exams** on week 6 and on week 12. Written Exam 1 (15%) and Written Exam 2 (15%) will count 30% in total. – 30%
6. *Students can choose to replace Written Exam 2 by submitting a Research Report based on their Research Proposal and presentation on week 12. – 15%*

AI policy

1. **Large language models (LLMs)** may be used to identify and define the research niche, refine the text of papers, and analyze data.
2. Students must provide **detailed information on how AI was used** in their work. When submitting written assignments, students must attach their **prompt history in a separate file**.
3. All assignments will be evaluated based on **creativity, links to the course material, sophistication, and quality of presentation**.
4. Students must have **full understanding and control of their work**. Inability to answer questions about the submitted work may result in a **reduction in points**.

Communication

- Materials:

https://github.com/blengyel/economic_complexity2026

- General communication:

Teams: CLOUD_SDS_26-26/Economic Complexity

- Personal communication:

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